



Teaching as a
Professional Discipline

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Chapter 1

The Paradigm Problem

There can be few subjects, if any, that experience as great a degree of internal dissension as education. All disciplines, of course, have their tensions, to do with their scope, theoretical persuasion, methodology, application or other aspects of their practice. Academic life is a disputatious affair, and the historical longevity of the university as an institution may be due partly to the fact that it has found ways of managing conflicts that would leave a church in schism, fragment a political party, or drive a commercial company to the wall.

The tensions within education, however, reflect the very existence of the discipline and nature of the activity. They lead some people to doubt whether education is a proper discipline at all, as distinct from a collection of elements borrowed from other disciplines. And these tensions have practical as well as academic consequences. They place a question mark against the idea of teaching as a profession and undermine its credibility as a policy voice. They make the field peculiarly vulnerable to external pressures. And they lead to the sudden lurches and longer-term pendulum swings that typify educational practice, making it a prey to passing fads and fashions.

Many of these tensions concern what is taught and what ought to be taught—the curriculum. This is hardly surprising. The curriculum involves assumptions about the purpose of education, the nature of knowledge, the social context and the development of the learner, all contentious areas. It is also subject to more immediate political pressures to ‘do something about’ current problems of a moral, social or economic kind. There is a very substantial body of curriculum theory that helps us to analyse these various perspectives, claims and tensions, but the curriculum in practice seems destined always to remain an arena of conflict, reflecting as it does the unresolved issues of the society in which it exists.

However, there are also major disagreements about the process of teaching, the how rather than the what of education, and it is with these that this book is concerned. Such tensions often take the form of arguments about methods: whether the class should be taught as a whole or in small groups, whether lectures are preferable to seminars, or what the proper role of practical work is. In recent years, the development of educational and communications technology has added an extra dimension to such debates.

I shall argue in this book that such arguments are not only unproductive, but a poor way of thinking about teaching. After all, no other profession conceives of its work in terms of methods in the way that

education does. Methods are involved in the practice of medicine, engineering and the law, but doctors, engineers and solicitors would not dream of reducing their work simply to this. Methods are means: they are ways of doing something, but they are not that something. One needs to look beyond methods and media to understand teaching.

This book will set out a multi-dimensional view of teaching in which methods and procedures constitute one, but only one dimension of the activity. The basis for this approach will be spelled out in Chapter 2, on The Nature of Professional Disciplines. I shall argue that all professional activities have certain basic characteristics, which suggest three basic questions: what do professionals do? what affects what they do? and how do they do it? The implications of this approach for teaching will be worked out first at the level of the course as a whole, in Chapter 3, and then at the more detailed level of the single class or teaching session, in Chapter 4. The two remaining chapters will explore the consequences of this approach for our ideas about professional expertise in general, and the training of teachers and evaluation of teaching in particular.

These ideas do not, of course, simply take vacant possession of an empty dwelling. The literature on teaching is rich in theories and models of teaching, and in addition practitioners bring with them their own personal and often implicit 'theories', which shape their practice (Marland, 1995). It is important to examine these first, not only to situate this approach in terms of current writing in the field, but to engage with what people already think and do.

One final point before we embark. I have referred only to 'teaching' rather than the 'teaching and learning' that seems to be de rigueur these days. Clearly, ideas about teaching must relate to ideas about learning, because the purpose of the former is in some way to enhance the latter, and there will be references to research on learning in various parts of the book. However, it is important to remember that learning goes on all the time, whether people are being taught or not. There is now widespread evidence from studies of adult learning (Brockett and Hiemstra, 1991; Candy, 1991; Gear, McIntosh and Squires, 1994; Tough, 1971) that the learning that goes on within the confines of formal education and training is only the visible tip of a much larger iceberg of ubiquitous, informal, self-directed learning. Learning happens anyway; it is teaching that is the special case, and it is for that reason that I shall focus on it, although as we shall see, the wheel will eventually come full circle.

Despite the fact that Masterman (1972) identified 21 different senses in which Kuhn (1962) used the word 'paradigm', it has entered academic currency as a useful term for describing not only the way in

which something is conceptualized or viewed, but the whole package of beliefs, values, attitudes and practices that goes along with that view (Burns, 1995). It will be used in that general sense here, to explore a number of different views of teaching, and the practical consequences of those views. Indeed the term is particularly apt in an educational context, because it accommodates some of the complexities of the relationship between abstract formulation, practical activity and cultural context that characterize the field.

For Kuhn, the task was to describe how one scientific paradigm came to displace another one, a process for which the Copernican revolution provided an exemplary case study. However, the problem in teaching is closer to what Masterman describes as 'multiple-paradigm science' (1972, p. 74). Here, as in some other social science fields, the problem is to explain not how one paradigm displaces another, but how a number of conflicting or competing paradigms somehow coexist. As Masterman notes, 'far from there being no paradigm, there are on the contrary too many' (ibid.). Thus I shall explore seven different paradigms of teaching that, while they may rise or fall in terms of relative influence, lay permanent claim to being good ways of thinking about teaching. They are: teaching as a common-sense activity, teaching as an art, teaching as a craft, teaching as an applied science, teaching as a system, teaching as reflective practice, and teaching as competence.

Each of these is an important paradigm, with a substantial literature attached, and it will be possible here only to sketch out the main features, strengths and limitations of each one. Each of them draws on sources of thinking that go well beyond education, an interesting fact in itself, but one that again will point up the limits of the analysis here. And the fact that each is treated discretely should not be taken to mean that there are no links and overlaps between them, or that one cannot draw on them eclectically, and find ways of accommodating elements of several of them. That would be to assume that teaching is an entirely homogeneous activity— that all teaching is somehow the same—and it may well be that different paradigms resonate with different teachers in relation to different aspects of their work.

Teaching as a Common-sense Activity

The idea that teaching is a common-sense activity is not one that is frequently heard in education faculties, for obvious reasons, but it is sometimes voiced by colleagues in other university departments. How else can one explain the fact that lecturers who have had no teacher training nevertheless produce their annual crop of graduates, and sometimes gain high teaching quality assessment ratings in the

process? Not only that: the common-sense paradigm has a perfectly respectable theoretical pedigree, and reflects the wider attachment to common sense that is a marked feature of English culture.

The common-sense paradigm of teaching rests on two different kinds of argument. The first goes back to the experience of schooling. Few things in society are actually mandatory, as distinct from forbidden. We do not have to vote; we do not have to get married; we do not have to join the army; we do not have to work (at least not yet). But we do have to go to school (or make alternative arrangements). This massive act of social compulsion, which is without parallel in terms of scope, length and intensity, means that we all have extensive experience of being taught, and (so the argument goes) when it comes to our turn to teach others, have some ideas about how to go about this, what to do and what not to do. The parts of teaching that we have to learn are those that are usually hidden from us as pupils or students, such as the administrative arrangements and the assessment process (which is typically wrapped in mystery).

The second basis of the common-sense argument is that the things we do when we teach are, by and large, not all that different from what we do in everyday life. We organize resources, plan events, explain things to people, ask and answer questions, guide, encourage and criticize. So all we need to learn when we begin to teach are those things that we do not normally do to others, such as lecture them or assess them, and we can pick these up through a process of trial and error or by apprenticing ourselves to a more experienced teacher. The other thing we do not normally do (with the exception of giving children's parties) is to cope with a room full of young people, and so classroom management is another skill that has to be acquired. But (so the argument goes) this can be picked up as well. Thus run the common-sense arguments for a common-sense approach to teaching, and they are sometimes applied to other 'non-technical' activities, such as management or social work, as well. And if teaching really is a matter of common sense, then it follows that most people can or could do it, since most people, though not all, have common sense (it is after all common rather than universal).

However, there is a more intellectualized version of this argument, which stems from the concept of common-sense knowledge. The central endeavour of phenomenology has been to explore not only how the world appears to us and how we experience it, but what cognitive or perceptual structures we bring to these appearances and experiences. In the work of Schutz (1970) and Berger and Luckmann (1971) in particular this has led to a preoccupation with the structures and characteristics of common-sense knowledge in the everyday 'lifeworld' (Lebenswelt). Such knowledge is described as pre-theoretical, in that it tends to take the world for granted rather than systematically analysing, questioning or theorizing about it—although 'a-theoretical' might be a better term, since common-

sense knowledge is not merely the precursor of organized knowledge, but exists in its own right. Common sense involves a certain stance vis-à-vis the world, which tends to accept and work with what is there. It is basically pragmatic or operational in its emphasis on actions and their consequences and places limits on the usefulness of analysis or reflection ('we don't want to get into all that'). It appeals to the shared wisdom of the tribe, although it allows for some individuality and innovation within those limits. And it is primarily concerned with know-how rather than knowledge; things are typically experienced in terms of their use (cf. Heidegger's notions of 'equipment' and 'tools') rather than as static objects of perception (Heidegger, 1962, pp. 95–9).

Another characteristic of common-sense knowledge is that despite the way it is sometimes phrased ('everyone knows that...') it does not aspire to strict generalization or universality; it tends to focus on the situation at hand. It is localized, specific, episodic and concrete. It addresses particular events, situations, problems. However, it is also associated with the notion of cumulative experience, and hence the idea that people can acquire it over time; young people typically lack it, and old people tend to lose it. One can thus see why it holds attractions for writers such as Hargreaves who conceptualizes teaching in terms of 'professional common-sense knowledge' (Hargreaves, 1993, pp. 86–92). Teaching, he argues, is localized, specific, episodic and concrete, so the knowledge we need to do it is largely of that kind.

One objection to Hargreaves' notion lies in the apparent contradiction between the concepts of profession and common sense. Professions are specialized, expert, closed, sometimes almost hieratic bodies, whereas common sense is by definition common property. Professional work seems to involve a range of kinds of expertise, including formal, organized knowledge and identifiable procedures and skills. We may use our common sense or judgment to synthesize and regulate these, but there has to be something to synthesize and regulate in the first place. Professional work may involve common sense, but it is hardly reducible to it.

A different kind of objection relates to the ways in which common-sense knowledge is stored and passed on, which is often through examples, cases and stories. The term 'anecdotal' used to be a term of mere abuse in the social sciences, but more recently the importance of anecdotes as ways of sharing and storing communal knowledge and know-how has become more widely recognized. An interesting question here is how far the current emphasis on narrative research methods in teaching (see the various articles in Gudmundsdottir, 1997) should in fact be located within a common-sense paradigm.

Anecdotes may play a particular role in specialized 'communities-of-practice', that is groups of people who are involved in the same kind of work, such as teaching (Brown and Duguid, 1996).

However, the problem with anecdotes is that one cannot disagree with them, as one might disagree with an argument. One cannot deny the story: one can only counter it with a different story. Thus anecdotal knowledge tends to carry a heavy normative charge, an implicit expectation about concurrence and assent. This makes it difficult for the practitioner to stand back from the practices of the group, and bring his or her analytical powers to bear. It can seem churlish to demur.

It was noted earlier that not only can the common-sense paradigm be defended in theoretical terms, but as something that reflects the wider society. The notion of common sense plays an important role in English-speaking and in particular English culture; England is, after all, the country of common land, common law, common prayer, a House of Commons. It would be interesting to explore the parallels with other countries: the French do not use *sens commun* much, and their *bon sens* appeals more to 'reason'; the German *Bauernverstand* has connotations of peasant wisdom or cunning; the Finns talk about 'countryman's sense'. In England, common sense seems to represent a permanent celebration of the demotic, a safeguard against ideological and other forms of intellectual extremism, a social bond that unites everyone from the prime minister down, perhaps a continuing inheritance from the old Anglo-Saxon culture with its moots and jury system where—at least in the historical mythology—the ordinary had its value and ordinary people had their say. And of course it serves to keep 'so-called experts' in their place.

However, it also sets limits to thought and analysis. To say that something is a matter of common sense is effectively to stop the debate, to rule out further discussion. Common sense tends to be conservative and reproductive in its instincts. It legitimizes a suspicion and ignorance of theory, abstraction, ideas, the intellect and the intellectual. It uses but debases the native English tradition of empiricism, of the appeal to experience. In what other country do people use the phrase 'purely academic' so dismissively, to mean of no real importance? In these respects, the common-sense paradigm may in the end impede our thinking about teaching and learning, as it does about much else.

Teaching as an Art

The idea that teaching is an art is both old-fashioned and quite recent. On the one hand, it is associated with some older books on the subject, such as Hightet's *The Art of Teaching*, which first appeared in 1951

and went through many reprints (Highet, 1963). In such accounts, teaching appears as a rather subtle and ultimately mysterious activity. It is strongly associated with personal qualities and characteristics; significantly Highet begins with a long section on 'The Teacher' and has a later section on 'Great Teachers and their Pupils'. While teaching involves methods, it also involves qualities—such as liking the subject and liking the pupils— and 'abilities' such as memory, will-power and kindness. Although Highet does not argue that teachers are born, not made, and indeed talks about 'teaching' in everyday life, there is a strong sense that selection or self-selection may be as important as training. Indeed, whereas the common-sense paradigm dismisses training as unnecessary, the art paradigm in its extreme form dismisses it as impossible. Teaching is a gift.

A somewhat different perspective on the art paradigm is offered by Axelrod (1973) in his book *The University Teacher as Artist*. Here, teaching is seen as the elaboration of style; style not as decorative adjunct or superficial appearance, but as the necessary and authentic expression of a way of doing or being. If the weakness of the art paradigm lies in its vagueness and openness to mere posturing—teaching as 'performance'—some of its truth surely lies in the notion of style as a coherent expression of identity. The teacher-as-artist does what he or she is.

As Delamont (1995) makes clear in her useful overview of subsequent thinking, the art paradigm has been marginal in most of the mainstream psychological and sociological literature on teaching. However, there are exceptions. Although Jackson does stress the importance of common sense (Jackson, 1986, pp. 10–23), his emphasis on the uncertainties of teaching and on its potential transformative role perhaps places him closer to the art paradigm than any other, although such pigeon-holing hardly does justice to his thinking. Delamont also mentions the work of Gage, which will be referred to later, but probably the most explicit formulation of the art paradigm is that of Eisner. He argues that teaching can be seen as an art in four senses: (1) that it is sometimes performed with such skill and grace that it can be described as an aesthetic experience, (2) that it involves qualitative judgements based on an unfolding course of action, (3) that it is contingent and unpredictable rather than routine, and (4) that its outcomes are often created in the process (Eisner, 1985, pp. 175–6). Tom, however, draws a distinction between the fine arts and practical arts, arguing that the latter provide a better metaphor for teaching, a point that will be picked up in the next section on craft (Tom, 1984, pp. 129–35).

However, beyond the work of these particular writers, there are reasons for thinking that the art paradigm has made something of a comeback in recent years. This is evident in subtle shifts in the discourse of research on teaching, and the increasing use of words such as 'artfulness', 'artistry',

‘repertoire’ and ‘improvisation’ in describing the process of teaching. The current interest in ‘narrative’, which was discussed earlier in relation to the common-sense paradigm, may also be linked to the art paradigm, although narrative is itself a problematic notion in a good deal of modern writing, and its use in educational research seems curiously old-fashioned. In one sense, these words constitute negative or residual elements: signs that rational, technical or scientific accounts have somehow failed to capture the process adequately (Dunkin, 1987b). However, they also signal a positive recognition of the indeterminacy of much teaching (Delamont, 1995) and, by extension, of the limitations of plans, scripts or procedures, the whole rational apparatus that occupies a good deal of teacher training.

The problem with the art paradigm lies as much in the art as the teaching. When teaching is said to be an art, it tends to be a rather romantic view of art that is implied, rather than the strict disciplines and rules of classicism. The emphasis is on expression and spontaneity, on creative responses to the unfolding situation, on personal authenticity and originality, rather than on more classical criteria such as good technique and sense of form. And the problem becomes more acute if one considers the contemporary arts. Whereas in the past one might identify a particular style or school of literature, painting or music as being the dominant and characteristic one in a particular period, we now seem to be in a situation that is not so much post-modernist as post-orthodox, with several different styles or traditions existing concurrently. To take just one example, there is contemporary poetry being written in relatively traditional modes, modernist poetry involving composition by field, using collage and found text, ‘language’ poetry that makes language itself the object of writing, performance poetry and concrete poetry, not to mention various other forms. And these are not just ‘styles’ in some superficial sense; they imply different views of literature, different notions of language, and different relationships with an audience, to the point where it is difficult to believe that we are talking about the same thing at all. If we cannot agree on what art is, how can we describe teaching as an art?

There may, however, be another kind of value in the art paradigm. Whatever the style or form of art, most artists would probably agree that the job is an internalized one. Learning to write, play or paint well involves a great deal of self-discipline, which may in fact be much more rigorous than any that might be imposed from the outside. In an era that places increasing emphasis on external controls—in terms of performance indicators, accountability, appraisal and the like—it is no bad thing for teachers to be reminded that the ultimate manager is oneself.

Teaching as a Craft

In contrast to the art paradigm, the craft paradigm is on the face of it much more straightforward. Whereas the arts are intuitive, personal, mysterious and typically the province of the gifted few, crafts are explicable, objective, transparent and accessible to the many, as the ubiquity of DIY attests. One can demonstrate a craft, imitate it, practise it, refine it and master it. The craft guilds are a major, if undervalued, strand in the history of European education and training. One of the tragedies of colonialism is that, with a few exceptions, the craft apprenticeship system was not exported along with the formal educational systems to the developing countries. After all, Europe developed historically on two educational legs, not one.

One can see the attractions of such a paradigm of professional work. If teaching is a craft, then one can analyse it into its elements, set up programmes to train people in it, and produce a large and skilled workforce for the schools, colleges and universities. There will be no mystery, no messing about, and it is perhaps no accident that the craft model seems strongest in that sector of the system that itself embodies such work, namely the vocational/technical Further Education colleges. The various City and Guilds teaching certificate courses are typical examples of their kind, and similarly practical courses exist for teachers and trainers in the German 'dual' system.

However, recent writing on the craft paradigm has gone well beyond the kind of straightforward practical approach adopted by Marland (1975) or, in the context of higher education, Eble (1976), although the latter also nods in the direction of the arts. In the hands of authors such as Desforges and McNamara (1979), Tom (1984), Tom and Valli (1990), Leinhardt (1990), Grimmett and MacKinnon (1992) and McNamara (1994) the concept has been extended and refined to the point where it is now a major contender for the dominant metaphor of teaching. Tom in particular adds a moral dimension to teaching as a craft, but the main thrust among all the writers is to develop a view of teaching know-how that goes beyond the general and formulaic, and attempts to capture the local, vernacular, specific reality of teaching as-it-is-done. As with the art paradigm, the craft view of teaching seems to be partly a reaction against the applied science paradigm, which views teaching in terms of the application of general

principles (see Tom, 1984, pp. 135–49). But teaching (the craft writers argue) is not general, it is always particular; it is not principled, it is fashioned; not abstract, but concrete.

Logically, if craft knowledge were wholly particular and always related to the specific case or situation, it would not accumulate and develop; it would remain trapped within its own specificity, never building up into anything more than individual responses to local situations. This is indeed a problem for those who stress the particularity of knowledge about teaching, for example in action research. What we learn about one situation may enable us to deal better with the situation in question, but if it does not generalize or transfer in some way to other situations, we can hardly talk about the development of expertise. And if we can generalize it, surely we can abstract it?

One response to this (see Clandinin and Connolly, 1996) is to argue that people can use such knowledge even when it cannot be generalized in the strict sense (i.e. without qualification) because they can recognize aspects of it and make the necessary translations and adaptations to their own case; it thus has affinities with ‘narrative’ in literature. We cannot experience the aristocratic society of nineteenth-century Russia as portrayed in *Anna Karenina*, but we can nevertheless project ourselves into it sufficiently to understand what the characters are going through. Craft knowledge, surprisingly perhaps, therefore implies a theory of the imagination: of the capacity to create cognitive scenarios based on, but not restricted to, what we know or have experienced directly. It is our imagination rather than our knowledge that allows us to identify or empathize with and learn from, for example, the experience of someone who teaches a different subject in a different school—in short, to enter the world of others, however imperfectly. In this way, it is the very concreteness of craft knowledge that provides the imaginative foundations for others to build on: the best case studies are not those that are most like our own situation, but those that are most real.

The second point to be made is that although researchers commonly refer to the problems of generalizing in education and other fields, they less frequently discuss the problem of particularizing. Writing about educational narratives, case studies or action research tends to assume that one can give a description of a particular situation, but in fact such particularities are themselves shot through with generalities in the very words they use. If a situation were wholly particular or unique we would not be able to describe it at all. Even an apparently specific episode such as ‘Peter came into the classroom, sat down at the back and started chatting to his friends. I told him to be quiet and get on with his work’ is riddled with generality. A ‘classroom’ is a general category of space; ‘told’ implies a kind of general authority; ‘work’ implies a general way of thinking about education. The point is simple, even banal:

however much we may think we are concentrating on the specifics of a situation, we are always enmeshed in the inherent generality of language and the shared structures of perception and cognition. The purely particular does not exist. Craft knowledge, like other kinds of knowledge, is embedded in something more general, although not necessarily generalizable in the strict scientific sense.

This concrete, grounded strength of the craft paradigm is, however, also its limitation. However much modern writers may try to extend its scope, its historical origins and connotations lie firmly in the artisan's fashioning of materials and objects. But students are not materials and learning is not an artefact. While the process may be equally non-formulaic in relation both to objects and people, the metaphor of craft in the end restricts what is an essentially human enterprise, concerned in the case of teaching and learning with meanings, not artefacts. Objects do not answer back; there is no possibility of dialogue with them; if we shape them wrongly, they do not protest. The value of the craft paradigm lies in illuminating the concrete aspects of professional work. It asserts the value of doing and experience, the endlessly repeated yet never quite similar encounters with the phenomenal world. It captures some of the grittiness of teaching, the grain of a particular group, the warp and weft of the process, the almost physical resistance of problems, the palpable satisfaction of a good outcome. But it is doubtful whether it can be a sufficient paradigm of teaching.

Teaching as an Applied Science

In its strong form, the applied science paradigm assumes that professional work involves the application of scientific principles and evidence to practical tasks. Science, it is argued, investigates and discovers fundamental patterns and regularities in the natural and social world. A knowledge of these patterns allows practitioners to intervene in events with a high degree of confidence that the effects of their interventions will produce the intended results. It is not necessary for such knowledge to be complete or such predictions to be certain; as long as the results are better than would otherwise be obtained, for example by relying on trial and error, common sense or intuition, the approach is justified. Thus the shift from a law-like conception of science in the early part of the century to a more probabilistic recent one does not affect this argument. It can be noted in passing that the idea that nature exemplifies 'laws' is itself a metaphor, and a characteristically European one, whereas the conception of orderliness in traditional Chinese science involved rather a sense of organic harmony (Needham, 1956, pp. 518–83).

Whereas the art and craft paradigms have long historical roots, the applied science paradigm depends on having some science to apply, and is thus a more modern idea. (The translation of Aristotle's *poiesis* as 'applied science' (Hope, 1952, p. 233) is a retrospective and dubious imposition of a contemporary concept.) As Gage (1996) argues, applied science has played a large part in some professions such as medicine and engineering, so why not in education? Thus, for much of this century, people looked to educational psychology in particular to provide a basis for teaching and learning, in terms of fundamental laws, principles and rules.

The applied science paradigm in its weaker form includes other 'foundation' disciplines such as philosophy, history and sociology, which are believed to provide a basis for practice, a belief that underpinned much teacher training in the 1960s and 1970s but came in for increasing criticism as time went on (see Hirst, 1983; Wilkin, 1996).

Most contemporary models of teaching are in fact psychology-based models. This is true of the majority of the models contained in the various editions of Joyce and Weil's *Models of Teaching* (1996) and the relevant sections in the two editions of the *International Encyclopaedia of Teaching and Teacher Education* edited respectively by Dunkin (1987a) and Anderson (1995). Some of these models derive directly from a particular school of learning theory. Thus one can see Skinner (1954) and Bruner (1968) as expressions of behavioural and cognitive theory respectively, and Rogers (1983) and Winne (1995) as representatives of humanistic and information-processing approaches; Bandura (1986) provides the basis for a theory of social learning and teaching through modelling. More recent models, such as those advanced by Entwistle (1992) and Ramsden (1992) also derive from a particular view of learning, in this case based heavily on Marton's work on student approaches to learning within the cognitive tradition (Marton, 1977). Interactive, mastery, heuristic and competence models of teaching all stem from particular psychological theories (see Dunkin, 1987a). Even where the model of teaching is more eclectic or generic, as in the case of Dunkin and Biddle (1974), Gagné (1985) or Biggs (1993), the debt to psychology is clear.

While psychology has lit up some of the darker corners of educational work, however, it has not provided the general illumination of the whole room which was expected earlier in the century (see for example Ward, 1926). We still do not have an applied science of teaching. Attempts to summarize the general implications of learning theories for teaching (see, for example, Hilgard and Bower, 1975, pp. 606–38) hardly add up to a coherent model. There are two problems here, related respectively to the science and the application. The fact that the human or social sciences have been less successful than the natural

sciences in generating high-confidence predictions is widely accepted. Two quite different explanations, however, are given for this shortfall. The first is that the human sciences are more complex in terms of the numbers and types of variables operating in human situations, and therefore require more complex methodological and statistical procedures to identify any patterns. Progress is therefore likely to be slow, and much slower than some of the pioneers in such fields originally expected, but it is being made. We are, it is argued, gradually building up a coherent understanding of human learning, for example through meta-analyses of large numbers of existing studies (Gage, 1996). Writers typically refer to such work as being in its infancy or at an early stage or having some way to go, but the assumption is that it will in fact retrace the developmental path of the natural sciences. It should be noted that this is simply a belief; despite the scientific nature of the work, there is no empirical basis for such an assumption.

The second explanation is that the existence of human consciousness introduces a qualitative difference between the non-human and human domains, because consciousness, in effect, breaks any patterns that we may identify. Even with something as basic as the hunger drive, people can choose to go on hunger strike to the point of death. Concepts such as 'reward' and 'punishment' are in fact highly subjective; some people find conventional rewards uninteresting, or conventional punishments rewarding. Culture can be defined precisely as that which is beyond nature, and there is no compelling reason to assume that it will exhibit patterns in the way that the natural world does. The problem is exemplified neatly in language, which is arguably the defining element of human culture. Language is both a patterned (syntactically) and unpredictable (semantically) phenomenon. Each language has its regularities, but each speech-act is unique. Attempts to reproduce the latter artificially run into Dreyfus's *ceteris paribus* problem: other things never are equal in the human domain (Dreyfus, 1992, pp. 57–61). The great twentieth-century project of trying to do in the human sphere, through economics, sociology and psychology, what has been done in the natural one runs apparently into the post-modernist wall: human beings are not only beyond nature, but beyond culture; consciousness, precisely because it is the capacity to think about thought, is the ultimate wild card.

This does not mean that the human sciences are useless, merely that we have to be more modest in what we expect of them. If a pattern exists 'for the most part', we can still base our decisions on it as long as we recognize that events may turn out otherwise. The human sciences can thus give us a broad steer in some aspects of professional work; but we need to realize that this is usually indicative rather than predictive. There may be rules, and even rules about rules, but in the end one is faced with a judgment as to whether to apply the rules. Thus the doctor will feed the results of, say, a 'scientific'

blood test into his or her decision-making, but the final treatment decision will have to take account of a wider range of considerations, including the general condition and situation of the patient, and the time-frame and availability of medical provision and intervention.

This leads to the second problem with the applied science paradigm, which concerns the application rather than the science. Part of the problem of relating the ‘foundation disciplines’ of education to the practice of teaching has been precisely the lack of a conceptual framework for the latter. There has been nothing, as it were, to plug the psychology or sociology into, hence it has been difficult for teachers to relate the generalities of such disciplines to the specifics of their classroom work. This has led to a good deal of criticism of the irrelevance of such disciplines, but in fact the problem lies not with them but with education itself. The failure of ‘educational theory’ in the 1960s and 1970s was not in fact a failure of educational theory, but of theories in education; there was little or no educational theory per se. One of the arguments for the model that will be set out in Chapters 3 and 4 is that it offers a framework to which philosophical, psychological and sociological content can be related.

Even if there has been less science to apply than some writers believe, what there has been has thus been under-utilized because of the absence of a coherent model of teaching that could take account of it. I shall argue in the next chapter that the contingent nature of professional work rules out, a priori, the idea that teaching, or any other profession, can be seen simply as an applied science. No professional domain—whether engineering, medicine, agriculture or any other—is merely the application of general scientific principles. But this does not exclude drawing on scientific knowledge as one element among others in practical decisions; it merely relegates its status from a determinant to an element.

Contemporary writers in the instructional science tradition (see Gage, 1996; Stones, 1994) take a much more sophisticated view of the applicability of science to teaching and learning than their predecessors, and indeed Gage’s views show an interesting shift over time. It is now widely acknowledged that whatever ‘fundamental pedagogical principles’ exist must be interpreted in the light of the situation, rather than simply or directly applied. This thus admits a degree of contingency, in the form of mediating factors, into the analysis. The difference between such a view and the approach to be presented here is that for the applied scientist such contingency is a complication, a problem that intrudes upon the application process, whereas here it is regarded as a defining characteristic of professional work.

Even if all other contingencies were somehow eradicated or controlled, there would still remain the fundamental contingency of time. Time is important in professional fields in a way that is quite different

from scientific ones; not for nothing is Heidegger's great phenomenological tract titled *Being and Time*. Time is the ever-present, inescapable consideration for the professional, whether doctor, lawyer or teacher. All that they do is done in time. Things happen in a sequence of events. The order of actions can have a crucial impact on the outcome. For the scientist, time also matters in the sense that experiments and investigations obviously occur in and over time; but their timing is not usually so crucial, because they do not form part of this inexorable serial flow. But it can matter greatly whether an illness is diagnosed in March or April, whether something happens at the beginning of a class or at the end, at what point a fertilizer is applied to a crop. How often the professional person says 'I wish I had done this a little earlier' or 'I wish I had waited'; how often they lament not having enough time. Thus, even if the human sciences did manage to generate strong general laws, those laws would still have to be applied in a temporal context, after something and before something else. In that sense, the 'timelessness' of science contrasts with the temporality of the professions.

Teaching as a System

The idea that professional work, such as teaching, can be seen in terms of a system is sometimes associated with the applied science paradigm, but actually has a quite different basis. Although some writers such as Checkland (1981) see the growth of systems theory as a response to the scientific problem of dealing with wholes rather than parts, systems thinking actually belongs to a rational rather than scientific tradition. (Schon's notion of 'technical rationality' unhelpfully conflates the two [Schon, 1983, 1987]). Systems thinking can be seen as one form of a more general rational or rationalistic paradigm, which has had a widespread influence on late twentieth-century thinking, particularly in the fields of human organization and public policy (Carley, 1980) and which has in turn generated concepts such as Herbert Simon's 'bounded rationality' (Simon, 1996). And although it would be a complex task to unravel the various historical antecedents of this, the main debt would seem to be not to Renaissance science but to the more general espousal of rationality and clarity as a means to human progress that came in later centuries, and is associated particularly with the Enlightenment. The distinction between ends and means, which is a cornerstone of rational analysis, is after all a philosophical rather than scientific one.

The modern concept of the system builds on these foundations (see Emery, 1969). The ends-means distinction remains essential, but tends to be expressed in the language of aims/goals/objectives on the one hand, and strategies/tactics/ resources on the other. However, modern systems theory adds three crucial elements: (1) the idea that the means-ends nexus can be seen as a whole, that is as a system, (2)

the idea that the system may itself form part of a hierarchy of supra-systems and sub-systems, and (3) that such systems may have the capacity to regulate themselves through feedback. The contribution of mathematics to modelling such systems has been important, though so far has had little impact on education. Thus what is sometimes referred to as a systems approach in education or training is based on a mixture of rational planning and systems theory. That theory developed mainly in the years following the Second World War, and as von Bertalanffy has argued, has a mechanistic strand, which developed out of the need to plan man-machine operations such as air defence, and an organismic strand, which has grown out of the study of biological systems (von Bertalanffy, 1968). Systems theory of both kinds introduced new concepts for thinking about intentional, organized human activities: notions such as synergy, dynamic interaction, equilibrium and equifinality. The work of Norbert Wiener (1948) on cybernetics contributed the crucial concept of feedback, which among other things offered a more elegant version of the behaviourists' 'knowledge of results'.

The concept of system is now pervasive in our culture, and has influenced the professions along with everything else. In education, it dovetailed neatly with the behaviourist emphasis on precise, observable and measurable objectives and outcomes, and step-by-step approaches (see Mager, 1962; Skinner, 1954; Tyler, 1949) so that the 'systems approach' to teaching is in fact a composite of three things: rational analysis, systems theory and behavioural psychology. This is seen as a technology that is viable in its own right and that does not have to be grounded in basic scientific laws or principles. The arguments are purely rational ones. How can we know if we have achieved our aims if we have not specified them clearly in the first place? How can we analyse the process of teaching and learning unless we break it down clearly into all its components? How can we design or implement anything if we do not go through the rational cycle of formulating objectives, deciding strategies, selecting resources, organizing activities, implementing delivery and evaluating the results? This is not the language of science, but of a particular form of rationality.

The main contribution of systems theory to education and training has been to help people to think about teaching as a complex whole (see Patrick, 1992, pp. 109–30; Richey, 1992; Romizowski, 1988). Systems thinking highlights the extent to which the various components of a system impact upon or interact with one another; it thus reduces the dangers of treating things in isolation. It also demands a thorough analysis of the sequences and stages involved in such an activity, so that one is less likely to miss things out. The typical systems diagram thus involves not only boxes and arrows to indicate the relationships between components, but flow-charts and decision-trees to set out the sequence of

events. Insofar as professional work involves the planning of complex wholes, such rationality has a place.

However, it is insufficient on its own, for two reasons. First, it is decontextualized, and does not take enough account of the contingencies of teaching and learning. It may look fine on paper, but in reality all sorts of factors break in upon the neat scheme: time pressures, organizational politics, people's attitudes, and the general confusion and uncertainty that in other circumstances have been called the 'fog of war'. Systems theorists sometimes distinguish between closed and open systems, the latter having more permeable boundaries and transactions with their environments. However, once that openness gets beyond a certain point, the very concept of a system begins to break down, since it implies a certain stability of structure and function.

Even in theory, however, rational planning may not be the optimum approach, because it is essentially abstracted from the people and contexts in which it has to operate. Lindblom (1959, 1979) has argued for a form of incrementalism in the public policy domain, mainly on the grounds that policy-making is better seen as a form of negotiation between competing interests than as a rational process. This essentially political model of planning in fact applies quite well sometimes to teaching: not only may there be negotiation between the teacher and his or her peers and organization, but within the classroom between teacher and learner. The capacity to engage in continual negotiation and therefore modification is, Lindblom argues, a better guarantee of effectiveness in an unpredictable or evolving situation than a pre-ordained rational plan. As noted earlier, improvisation and spontaneity may therefore have their place in teaching along with preparation and planning.

The other basic problem with the systems approach is that it is purely a process model; it says nothing about the content of any system. It can be applied equally to firing a missile, running a small business, organizing health care or teaching primary school children. This is both its strength and its weakness. It is a powerful, generic idea, but tells us nothing about the substance of the particular activity we apply it to. It does not say anything in itself about the uniqueness of teaching and learning, and therefore cannot take account of the nature of those particular processes. It assumes that teaching and learning are clear, transparent, goal-directed, analysable, plannable and measurable activities, but it does not attempt to justify such assumptions. It has nothing to say about the functions of teaching, what teachers do. For these reasons, it is not a paradigm of teaching at all, but simply a general paradigm that happens to be applied to teaching, among other fields.

Teaching as Reflective Practice

Another example of a process paradigm is the notion of 'reflective practice'. This idea is associated in particular with the work of Schon (1983, 1987), who has related it to a number of professions (such as management and architecture) including teaching. It has also had considerable influence on other fields such as nursing and social work. Schon's work (together with that of Argyris, 1982) can be seen as part of a general reaction against 'technical rationality' and the beliefs, set out above, that professional practice could be based either on scientific principles or rational planning (although as noted earlier, their reaction tends not to distinguish between the two). The whole anti-technicist approach can be mapped neatly onto Kolb's experiential learning cycle, with its four stages of concrete experience, reflective observation, abstract conceptualization and active experimentation (Kolb, 1984). Critics of applied science or systems approaches to teaching variously place the emphasis on experiential learning (Boud, Cohen and Walker 1993; Weil and McGill, 1989); reflective practice (Calderhead and Gates, 1993; Schon, 1987); or action learning and research (Elliott, 1993; McGill and Beaty, 1992; Revans, 1982). The fourth stage—abstract conceptualization—receives much less emphasis in this paradigm.

Schon argues that most professional problems are messy and inchoate, and cannot be solved purely through scientific application or rational analysis. They require a continuing process of reflection on what is happening and has happened (reflection in action, reflection on action). Such a process can improve the quality of decisions, and is contrasted with unreflective, unthinking, routine or habit-ridden behaviour. Eraut (1995) sees Schon's approach as a form of metacognition and Argyris talks about 'double loop learning', which involves questioning the frame of reference of the question (Argyris, 1994). Thus practitioners have to disengage temporarily from the immediacy of practice to think about what they are doing—and about what they are thinking about it.

There can be no objection to the idea of reflective practice in itself. As noted earlier, it is difficult to imagine any professional activity going on without an element of reflection, if it is not to become wholly routinized and habit-ridden. The need for reflection is a direct consequence of the fact that professional work is contingent; if it were simply algorithmic and rule-governed we would not need to reflect on it at all. And one can see in practical terms what Schon is talking about. The advanced driver who is told to commentate aloud on his perceptions and decisions as he is driving (reflection-in-action) and the air force pilot who is carefully debriefed after a mission (reflection-on-action) offer two examples from quite

different domains, but one can see the potential use of such commentaries and debriefings in professional training.

Again, however, there are problems. If reflection is taken to be the only or main element in professional activity, it can lead to an underemphasis on sub-stantive knowledge or relevant skills. To what extent can one reflect without an adequate knowledge-base in the field? What is one reflecting with or about? And does one have the skills needed to translate the outcomes of such reflection into action? It is all very well for a teacher to be able to produce a reflective account of a class or session, but that is not an end in itself; reflection has ultimately to feed back into action.

Moreover, the relationship between reasoning and reflection is also not clear. The latter implies something more heuristic or iterative and less logical or systematic than the former. So how do we evaluate the quality of reflection? We can apply certain criteria to reasoning (to do with the grasp of concepts, process of argument and handling of evidence) but it is not so clear how we should judge a reflective account, such as a diary, journal or logbook. When we use words like 'thoughtful', 'perceptive', 'sensitive' or 'illuminating', what kinds of judgments are we making? Presumably the adequacy of a piece of reflection has to relate not simply to the person who is doing it, and how he or she has done it, but to the situation, and this sends us back to the thorny question of what is significant. If I reflect on the colour of the ceiling in my classroom, how do we know that that is not significant (if it isn't)? On what basis do we select for reflection and analysis a relatively small number of features from the infinite detail of the class? As Albert Camus once pointed out, true naturalism in writing is an impossibility.

As noted above, reflective practice is another pure process paradigm; it tells us nothing specific about teaching or any other profession to which it relates. So what do we reflect on and with? The answer presumably is 'experience' or 'practice'; but here again one confronts the problem for the empiricist: experience and practice are not some kind of raw material mutely awaiting interpretation, they are already 'structured' and 'processed' to some degree. Cognition is already deeply implicated in perception and experience; we do not perceive or experience first and think second.

While the notion of reflective practice is undoubtedly useful as an element in professional work and training, it is thus not enough on its own. It leaves unanswered questions about the content of such reflection. It is often seen as the opposite of systems thinking, but in fact it shares with that paradigm the strength and weakness of a process paradigm, applicable to everything and nothing in particular. One can also ask questions about the intellectual context of its use. The historical salience of science and

technology and the related cult of the expert in American culture make it understandable that there should be periodic reactions against this, and Schon fits neatly into this pattern. He can be seen as a scion of American pragmatism; there are affinities with Dewey (on whom he did his thesis) for example, in the emphasis on experience and inquiry, a debt which he has acknowledged (Argyris and Schon, 1996, pp. 30–7). However, the impact and reception of Schon in England is another matter. This is a culture in which science and technology have long had an ambiguous status, in which experts are commonly mistrusted, in which theory and abstraction are often regarded as alien. The notion of reflection fits comfortably into a certain amateur (rather than expert) quasiliterary and anti- technological ethos, which has deep social and cultural roots. It is hardly surprising, therefore, that (as in the Gospel) the prophet has been so much honoured in this country, but also rather worrying.

Teaching as Competence

The final paradigm is that which views teaching as a matter of competence or competences (the word ‘competencies’ is also sometimes used, but there seems to be no consistent difference in meaning). The development of competence-based education and training in the US has been traced by Tuxworth (1982), Short (1984) and Houston (1987). The approach there developed mainly in the fields of vocational education and teacher education in the 1960s and 1970s, although as with most fundamental educational ideas one can find precursors, in the objectives movement of the 1950s (Mager, 1962; Tyler, 1949) and even the efficiency movement of the 1920s (Bobbitt, 1924; Charters, 1929). Like these two earlier examples, the competence movement focuses on what professionals can actually do rather than what they know. The bottom line is the capacity of the practitioner to perform or deliver in the real-world context, a context that may bring pressures of time, uncertainty, conflicting priorities and inadequate resources. Competence shares with behaviourism the emphasis on specific objectives and observable behaviour, but goes beyond the earlier movement in deriving those behaviours from, and relating them to, the actual work environment.

In the UK the competence movement essentially dates from the formation of the National Council for Vocational Qualifications in the mid-1980s, and the espousal by that organization of a competence-based approach (de Ville, 1986). This has led to the reformulation of most vocational and technical courses in terms of a framework of competences and levels (usually 1 to 4) although the relationship between the NCVQ (now Qualifications and Curriculum Authority) and the existing examining bodies

such as the Business/Technician Education Council (BTEC) and City and Guilds was by no means always smooth. This was hardly surprising: such bodies were well established in their own right, and some of them date back to the nineteenth century. However, the NCVQ had powerful government backing, and (despite some practical criticisms) a good deal of support from employers, so the changes largely went through. These changes can best be seen not as changes in the content of the curriculum but as a re-formatting of courses in a way that emphasizes student performance and outcomes, and the assessment of these in the real work environment (where possible).

The extension of the paradigm to higher-level professional courses has been much more problematic. In some fields, such as education, nursing and social work, the competence-based approach has had considerable influence, though not usually in as detailed, specific or behavioural a form as in vocational and technical courses. Both Elliott (1991, pp. 118–134) and Barnett (1994, pp. 157–171) have argued for broader or more holistic approaches to competence in education, and in the ASSET programme for framing social workers, specific competences are complemented by an analysis of the general qualities required by professionals in the field (Winter and Maisch, 1996, pp. 39–63). Such work offers an interesting though currently less influential alternative to the dominant behavioural paradigm. Other professions, such as medicine, law and engineering, have so far largely resisted such ideas, partly because they are deemed inappropriate to the field, and also probably because they are viewed as an encroachment on their autonomy by an external body.

The competence paradigm in this country is well represented by Burke (1989, 1995) and Jessup (1991), who are largely sympathetic to it. Eraut (1994) has engaged dispassionately with it, placing it in the wider context of research on expertise; the most forceful of many critics has probably been Hyland (1997). There is also now quite a large practical literature on the subject (see, for example, Fletcher, 1991). At first sight, the competence paradigm might seem very similar to the craft one, in its emphasis on practical skill. However, the notion of competence goes beyond the connotations of craft skills, since it includes everything that goes towards the delivery or performance of those skills in the real-world situation, including, for example, the appropriate matching of skill to task, and the attitudes and stamina needed to carry it through in difficult circumstances. On the other hand, 'competence' has a narrower or more definable sense than 'art', in that it implies that what people do and the way they do it can be specified in quite clear and predictable terms. Although the levels of competence (from 1 to 5) defined by the NCVQ imply increasing open-endedness and contingency in tasks and situations, the general thrust behind the competence movement is to pin down the idea of doing.

The potential advantages of this approach are threefold. In general terms, it places the emphasis in vocational, technical and professional fields firmly where it should be: on performance, rather than knowledge or understanding. The bottom line is what the practitioner can do, rather than what he or she can describe, analyse, reflect on or explain. In terms of the process of education and training, a competence-based approach offers clear objectives for teacher and learners alike, thus clearing away some of the curricular fog that obscures courses, and providing an explicit, shared basis for working together. Thirdly, in terms of learning and assessment, it allows one to disaggregate large blocks of learning into smaller more manageable units, and offers a way of accrediting learning that has taken place elsewhere, on other courses, in other modes or through prior experience. In many ways, therefore, it promises to free up education and training from some of its supply-side institutional and curricular rigidities, and to shift the emphasis from producer to consumer. Its advocates certainly see it as not merely a technical approach but a liberating movement.

Many of the criticisms have concerned the implementation of the approach, and the unwieldy jargon and bureaucracy that it seems to engender. Some critics argue that mere competence encourages mediocrity, and that one should be aiming at excellence, or at least allow for a range of performance, not just the criterion-referenced tick/cross. More serious charges have been laid at the unreliability of assessment, much of which has to be devolved to practitioner-assessors in work environments who may apply varying standards in their local contexts, and this has undermined the credibility of some courses. To be fair, this is a problem with any work-based assessment, and a potential weakness even in well-organized apprenticeship systems such as the German one.

However, the basic weakness of the competence paradigm lies not in the process of teaching and assessment itself, but in the prior analysis of jobs that has to be carried out in order to identify competence. Such 'functional analysis', as it is called, has some affinities with the methods of task analysis developed in the 1960s and 1970s (see Stammers and Patrick, 1975), which essentially involve a hierarchical breakdown or 'progressive desegregation' (Ellis, 1995) of the work to be performed. The terms used for this vary, but typically a whole occupation or job is divided into key elements, which are in turn subdivided into tasks, which are in turn broken down into behaviours, which in turn generate specific performance criteria. A familiar tree-like structure thus emerges, though this is not necessarily symmetrical, and the number of levels can vary. This analysis is often complemented by a specification of the knowledge and understanding that 'underpin' the competences, and a list of the range of situations or cases to which they should apply.

The problem is that although the competence approach may look logical on paper, there is no reason to believe that it is necessarily a good way of capturing what people do in their work. There are two basic issues. The first is that in all but the simplest jobs or tasks, one has to distinguish between what one does and how one does it, and the relationship between these is not simply a hierarchical one. In the model set out in Chapter 3, I shall distinguish between functions or roles on the one hand (what one does) and procedures or methods on the other (how one does it). The point may be clarified by taking an example from outside the field of education.

Let us assume that one of the functions of a doctor is to diagnose a medical problem, that is to come up with a plausible relationship between the symptom the patient is showing (tiredness) and the possible cause or causes of this (stress, diabetes, heart problems, and so on). We might assume that the doctor arrives at a diagnosis primarily through using verbal procedures (question and answer, talk) and physical procedures (physical examination). However, the doctor may also on occasions use biochemical procedures (blood or urine tests) and even surgical procedures (taking a sample of tissue). So the single function of diagnosis (what) may be accomplished by the use of multiple procedures (how). Conversely, each of these procedures may serve multiple functions. Surgery is usually associated with treatment rather than diagnosis, as are biochemical procedures such as putting the patient on a drip or giving him or her a blood transfusion. Physical procedures (manipulation, massage) may also be curative rather than diagnostic and doctors' talk may serve a wide range of purposes beyond the diagnostic, ranging from treatment (persuading the patient to take the whole course of pills, therapy) through counselling to education. The relationship between what doctors do (functions) and how they do it (procedures) is thus not hierarchical but orthogonal. And the same applies to other professions.

Another way of putting this is in terms of that little word 'do'. For all its apparent concreteness or obviousness, the word do is in fact quite problematic. 'I am preventing the spread of disease' and 'I am giving an injection' are both possible answers to the question: what are you doing? The competence approach sees this difference as one of levels (disease prevention is a general task that can be subdivided into more specific behaviours). But there are many different ways of preventing the spread of diseases (sanitation, isolation, education, and so on) and there are many different functions of injections (to inoculate, give antibiotics, supply vitamins, control diabetes, sedate, and so on). The relationship between what one does and how one does it is very complex in professional work and cannot be represented simply in terms of hierarchical levels.

The second main problem with the competence paradigm is the way it deals with contextuality and contingency. As we have seen, the competence approach tries to accommodate contextual variations through 'range statements', that is the variety of situations in which the person should be able to perform. This implies that the competence is a standard behaviour that has to be adapted or fine-tuned to the context. The problem is that contingency is not simply an adaptation of, but a basic characteristic of, professional work. The world of the professional practitioner—the doctor, engineer, social worker, teacher—is a contingent world because it is not simply or normally a rule-governed world. That is why doctors and other professionals talk so much about professional judgment, good judgment and errors of judgment. Context is a permanent aspect of what they do; it defines one dimension of their world. It is not some kind of extraneous consideration to be tacked on in the form of a range statement.

The degree of contingency is sometimes seen as the key difference between 'low-level' vocational/technical occupations and 'high-level' professional ones; hence the common argument that the competence approach is suitable for more routine, simpler tasks, but decreasingly appropriate as one moves up the occupational ladder. However, one might argue that even the simplest and most routine task is subject to some contingencies, if only those of time (normal, rushed, panicstations), space (adequate, roomy, cramped) and resources (generous, inadequate, shoe-string). This points to the need for yet another dimension in any model of work, and it seems more logical to apply this to all occupations and tasks at all levels, recognizing that simpler jobs will produce simpler models, but still take the same basic form.

Conclusion

It would be a complex task, and one that would take us away from the main purpose of this book, to trace the influence of these seven paradigms on the various sectors of education at various times and in various places. Besides, they do not usually manifest themselves in pure or discrete form, but rather in the messy, semiconscious and eclectic use that characterizes much of our practice. We may espouse one paradigm, and act out another in our work; or assent to one in public, while holding to another in private. We may shift from one to another over time, perhaps in the developmental sequence suggested by Pratt (1989). And, as noted at the beginning, different paradigms may reflect different aspects of our work.

The same problem arises at the level of national policies and trends. It is one thing to describe, as Wilkin (1996) does, how the 'foundations' approach to teacher education in the 1960s and 1970s has largely given way to, on the one hand, a government-backed competence approach, and on the other, an emphasis on reflective practice in many university education faculties. But this is to speak only of the schools. Further education has hitherto been strongly influenced by the craft paradigm (in the various City and Guilds courses), which is not surprising in a sector that is heavily involved in teaching crafts of various kinds. In the field of training and development, the systems approach still largely rules, and again this is congruent with much of the language and ethos of management, with its emphasis on objectives, strategies, resources and evaluation. In higher education, the training of lecturers is a recent development, which shows various competing influences at the time of writing, and which will no doubt encounter some eloquent defenders of the common-sense and artistic paradigms.

The analysis sketched out here will, however, have served its purpose if it suggests why none of these paradigms quite fits the bill. Each one has its strengths, its truths; one cannot dismiss any of them out of hand. Indeed, a useful exercise is to ask practitioners to place the seven in order of preference. In my experience, this usually yields a wide diversity of responses and promotes lively discussion as people justify their preferences to one another. (I have yet to find two people in a group coming up with exactly the same ranking). Such an exercise helps to bring to the surface and engage the kind of tacit or hidden 'personal theories' which, as noted at the outset, teachers and intending teachers often bring to their work.

But each paradigm has its limitations. If teaching really were a matter of common sense, why do we find bad or incompetent teaching? If it is an art, why is it a (relatively) mass activity? Can an inanimate metaphor like craft really illuminate a human process? How can it be an applied science when there is not much science to apply? How can the messiness of teaching be accommodated in the notion of a system? What do we reflect with?

The next chapter will set out yet another paradigm, that of teaching as a professional activity. Again, this will prove to have its limitations, but I shall argue that it offers a more comprehensive and balanced view of teaching than any of the ones reviewed in this chapter. It will also help to relate teaching to other professions, bringing to light some interesting similarities and differences. But first we have to go back, as it were, to square one.

